

Source Corrections in Examples, More on Algebra and Brauer Groups

Complete Editorial Arguments

AI editorial work: OpenAI Codex—GPT-6 Astra, Ultra effort

30 September 2026

The original source is by the Stacks Project authors. This supplement proves five identified source corrections and their exact propagation. The original chapters accompany it with thirteen bounded edits. The longer arguments and stronger conclusions below are editorial material; they do not replace the source exposition. No human review, novelty, official Stacks endorsement, or resolution of the separately received open Brauer problem is claimed. The original proof notes and source locators remain included in the editable package. Their preparation-stage status statements are historical; release transitions are recorded separately.

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1 Three source corrections received from the Algebra review

These arguments concern three identified findings in two other chapters. They do not represent a complete review of either chapter. The proposed source edits consist of one vector index, two tensor subscripts, and one pair of parentheses. Complete arguments and their propagation remain identified editorial material.

The original source is by the Stacks Project authors, at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The earlier receiving records are ALGEBRA-RECON-1069, ALGEBRA-RECON-997, and ALGEBRA-RECON-1333, respectively. This note supplies the exact receiving calculations. No novelty, independent human review, or formal proof certification is claimed.

CC-FITTING-INDEX: the appended kernel vectors

Source: More on Algebra, Fitting ideals, lines 1355–1411. The erroneous index is at line 1385.

Keep the source ring R , its finite module M , a surjection $\pi : R^{\oplus n} \rightarrow M$, and its kernel K . Fix $0 \leq k \leq n$ and put $q = n - k$, only as an additional name for that integer. For chosen vectors $z_1, \dots, z_{n-k} \in K$, retain their full coordinate matrix

$$Z = (z_{ij})_{1 \leq i \leq n, 1 \leq j \leq n-k}.$$

After adding n' generators that map to zero, the surjection is

$$\pi' : R^{\oplus n} \oplus R^{\oplus n'} \rightarrow M, \quad (a, b) \mapsto \pi(a), \quad K' = K \oplus R^{\oplus n'}.$$

The required list has exactly $n + n' - k$ vectors:

$$z'_j = (z_j, 0) \quad (1 \leq j \leq n - k), \quad z'_{n-k+j'} = (0, e_{j'}) \quad (1 \leq j' \leq n'), \quad (\text{F1})$$

where $e_{j'}$ is the j' -th standard coordinate vector of $R^{\oplus n'}$. Its coordinate matrix is precisely

$$Z' = \begin{pmatrix} Z & 0_{n \times n'} \\ 0_{n' \times (n-k)} & 1_{n'} \end{pmatrix}. \quad (\text{F2})$$

The printed index $n + j'$ leaves the positions $n - k + 1, \dots, n$ undefined when $k > 0$, and its largest indices exceed the stated list. For example $n = 2, k = 1, n' = 1$ requires two columns numbered 1 and 2, whereas the printed rule calls the added column 3. Equation (F1) repairs exactly this index; neither the vector nor its coordinates changes.

We prove the ideal comparison used by the source. Every $(n + n' - k)$ -row minor of (F2) uses all its columns. A selected row set that omits a bottom row makes the corresponding added coordinate column zero, so its determinant is zero. A selected row set containing every bottom row has $n - k$ top rows. With both rows and columns in their original increasing order, its determinant equals the corresponding $(n - k)$ -row minor of Z : its matrix is block diagonal with bottom block $1_{n'}$. Every such minor of Z occurs. This proves equality of these two minor ideals, including $n' = 0$ and $k = n$. In the latter case the empty top determinant is 1, and the bottom identity matrix provides the unit minor.

Conversely choose any $n + n' - k$ vectors in K' , with coordinate matrix W , and project each of them to K . Write their projected matrix P and their bottom coordinate matrix Q , so

$$W = \begin{pmatrix} P \\ Q \end{pmatrix}, \quad P \in \text{Mat}_{n \times (n+n'-k)}(R), \quad Q \in \text{Mat}_{n' \times (n+n'-k)}(R).$$

For a selected $(n + n' - k)$ -row minor, let s be the number of selected bottom rows. Expansion along these s rows is the sum over all s -element column subsets of the selected bottom determinant times the complementary $(n + n' - k - s)$ -row determinant of P , with its row-selection and column-selection sign retained. Since $s \leq n'$, the latter determinant has size at least $n - k$. Expanding it along any $n - k$ selected rows writes it as a signed sum of $(n - k)$ -row minors of P multiplied by complementary minors. Each of its columns is one of the original projected vectors in K . Every term therefore belongs to the source ideal generated by the $(n - k)$ -row minors of kernel vectors. For $k = n$ this containment is containment in the unit ideal. This proves the other inclusion without discarding any sign or coordinate contribution.

For completeness, this kernel-vector description agrees with the presentation matrix description even if the relation index set is infinite: every kernel vector is a finite linear combination of its columns. Multilinearity expands a chosen determinant into a finite sum of coefficient products times minors of the presentation matrix. Terms with repeated columns vanish by alternation. Conversely, each presentation column is itself a kernel vector, so every presentation minor occurs in the kernel-vector ideal.

The comparison between different generating surjections has an explicit ambient isomorphism. For $\rho : R^{\oplus m} \rightarrow M$, choose lifts

$$U : R^{\oplus m} \rightarrow R^{\oplus n}, \quad \pi U = \rho, \quad V : R^{\oplus n} \rightarrow R^{\oplus m}, \quad \rho V = \pi.$$

These exist by choosing a lift of the image of each standard basis vector. On $R^{\oplus n} \oplus R^{\oplus m}$, the automorphisms

$$\begin{aligned} T_\pi(a, b) &= (a + Ub, b), & T_\pi^{-1}(a, b) &= (a - Ub, b), \\ T_\rho(a, b) &= (a, b + Va), & T_\rho^{-1}(a, b) &= (a, b - Va) \end{aligned}$$

satisfy

$$(\pi, 0)T_\pi = (\pi, \rho) = (0, \rho)T_\rho.$$

Thus $T_\rho T_\pi^{-1}$ identifies $\ker(\pi) \oplus R^{\oplus m}$ with $R^{\oplus n} \oplus \ker(\rho)$, and both composites with the surjections are the displayed maps to the original M . The minors of a matrix multiplied by an ambient automorphism belong to the original minor ideal by the determinant expansion. Multiplication by the inverse proves the reverse inclusion. A permutation of the two coordinate blocks changes only the corresponding determinant signs, which are units and remain present in the expansion. This proves the independence comparison with the corrected indices.

For $k > n$, the k -th Fitting ideal for a presentation with n generators is the unit ideal. This convention is consistent with stabilization: if $0 < n + n' - k \leq n'$, select that many added coordinate vectors and the same bottom coordinates to obtain determinant 1; if $n + n' - k \leq 0$, the unit convention applies directly. No claim about a negative number of chosen kernel vectors is needed.

Propagation: the Jacobian-defect calculations in the receiving Algebra note use presentation independence. Their matrices and Fitting ideals are unchanged; equation (F1) gives the correctly indexed comparison. The source theorem is retained, and the longer verification stays in this editorial note.

CC-CONORMAL-BASE: the tensor product is over B

Source: More on Algebra, the transitivity lemma, lines 8458–8505. The wrong tensor subscripts occur at lines 8502 and 8504.

Retain all the source maps

$$A \rightarrow B \rightarrow C, \quad P = A[x_s \mid s \in S] \twoheadrightarrow B = P/I, \quad B[y_1, \dots, y_m] \twoheadrightarrow C.$$

Put $T = P[y_1, \dots, y_m]$, let $K = \ker(T \rightarrow C)$, and let $J = \ker(B[y_1, \dots, y_m] \rightarrow C) = K/IT$. Here IK means the ideal $(IT)K$ of T , as in the source. The B -action on I/I^2 is induced by the original quotient $P \rightarrow P/I$; the B -action on C is the original given map.

For these data, without an additional regularity assumption, the map

$$\Phi : (I/I^2) \otimes_B C \xrightarrow{\sim} IT/IK, \quad (i \bmod I^2) \otimes (t \bmod K) \mapsto it \bmod IK \quad (\text{C1})$$

is an isomorphism of C -modules. If i changes by an element of I^2 , its product with t changes by an element of $I^2T \subset IK$, since $IT \subset K$. If t changes by an element of K , its product with i changes by an element of IK . For a lift $p \in P$ of $b \in B$, the two balanced tensors $(ip \bmod I^2) \otimes \bar{t}$ and $(i \bmod I^2) \otimes \overline{pt}$ have the same image. A different lift of b makes their products differ by an element of I^2T . This proves every required well-definedness and balance relation.

Here is an inverse retaining every polynomial coefficient. Each $q \in IT$ has a unique finite polynomial expansion

$$q = \sum_{v \in \mathbb{N}^m} i_v y^v, \quad i_v \in I,$$

and define

$$\Psi(q \bmod IK) = \sum_{v \in \mathbb{N}^m} (i_v \bmod I^2) \otimes \overline{y^v}. \quad (\text{C2})$$

The sum is finite. Coefficient multiplication by $p \in P$ is balanced with its original image in B ; multiplication by y_t is balanced with its image in C . Consequently the expression for $q = i h$, with $i \in I, h \in T$, is $(i \bmod I^2) \otimes \bar{h}$. Every element of IK is a finite sum of terms $i h k$, where $k \in K$; each maps to $(i \bmod I^2) \otimes \bar{h}k = 0$. Thus (C2) descends to the quotient. Applying (C1) to (C2) returns the complete polynomial q modulo IK . Applying (C2) to (C1) on an elementary tensor returns that tensor by expanding t into all its polynomial coefficients. Both composites are identities, and the displayed balance proves C -linearity.

The source additionally assumes $B \rightarrow C$ is a local complete intersection. We verify the exact consequence it needs:

$$IT \cap K^2 = IK. \quad (\text{C3})$$

The source definition makes J a Koszul-regular ideal, hence locally generated by a sequence whose first Koszul homology vanishes. For the relevant definition and original proofs see regular ideals, lines 8100–8137, the intersection calculation, lines 7471–7520, and its ideal form, lines 8234–8244.

The needed calculation is as follows. Let D be a ring, L an ideal, and $g_1, \dots, g_r \in D$ whose images in D/L have vanishing first Koszul homology. For $h = \sum_i g_i a_i \in L$, the vector $\sum_i \bar{a}_i e_i$ is a Koszul cycle. With the convention

$$d_2(e_i \wedge e_j) = \bar{g}_i e_j - \bar{g}_j e_i \quad (i < j),$$

there are coefficients $\bar{c}_{ij} \in D/L$ such that it equals $\sum_{i < j} \bar{c}_{ij}(\bar{g}_i e_j - \bar{g}_j e_i)$. Choose lifts $c_{ij} \in D$ and set

$$\ell_i = a_i + \sum_{j > i} c_{ij} g_j - \sum_{h < i} c_{hi} g_h \in L.$$

Then, keeping both signed terms for every pair,

$$\sum_i g_i a_i = \sum_i g_i \ell_i + \sum_{i < j} (-c_{ij} g_i g_j + c_{ij} g_j g_i) = \sum_i g_i \ell_i.$$

This proves $L \cap (g_1, \dots, g_r) = L(g_1, \dots, g_r)$; the other containment follows by multiplication. If $Q = L + (g_1, \dots, g_r)$, an element of $L \cap Q^2$ has the form

$$h = \sum_{i,j} b_{ij} g_i g_j + \sum_i \ell'_i g_i + \ell', \quad b_{ij} \in D, \quad \ell'_i \in L, \quad \ell' \in L^2.$$

The first sum belongs to $L \cap (g_1, \dots, g_r)$, and is therefore in $L(g_1, \dots, g_r)$. Every displayed summand lies in LQ . Hence $L \cap Q^2 = LQ$, including the empty-sequence case.

Apply this with the actual localized ring $D = T_g$, $L = (IT)_g$, and $Q = K_g$ around each prime of T containing K . Such a g is the lift of a source neighborhood on which J has Koszul-regular generators. At a prime not containing K , K localizes to the unit ideal and both sides of (C3) localize to IT . Localization commutes with these products and this finite intersection: an element in both localized ideals has two denominator witnesses, and their product is a common witness. Thus the quotient $(IT \cap K^2)/IK$ vanishes at every prime. If this module had a nonzero element, a maximal ideal containing its annihilator would give a localization in which it survives. This proves (C3) globally. No finite generation of I, K , or that quotient was assumed.

By (C1) and (C3), the natural map

$$(I/I^2) \otimes_B C \rightarrow K/K^2, \quad \bar{i} \otimes \bar{t} \mapsto it \bmod K^2$$

is injective. The map $K/K^2 \rightarrow J/J^2$ is surjective, and its kernel is $(IT + K^2)/K^2$, identified with $IT/(IT \cap K^2)$. Therefore the exact row in the source is precisely

$$0 \rightarrow (I/I^2) \otimes_B C \rightarrow K/K^2 \rightarrow J/J^2 \rightarrow 0. \quad (\text{C4})$$

The polynomial differential row retains the separate dx_s and dy_t coordinates:

$$0 \rightarrow \bigoplus_{s \in S} C dx_s \rightarrow \left(\bigoplus_{s \in S} C dx_s \right) \oplus \left(\bigoplus_{t=1}^m C dy_t \right) \rightarrow \bigoplus_{t=1}^m C dy_t \rightarrow 0. \quad (C5)$$

The conormal differentials send an equation to its original universal derivative tensored with C . For $i \in I, t \in T$, the identity $d(it) = t di + i dt$ becomes $t di$, because i maps to zero in C . This proves commutation at the left; reducing an equation modulo I and taking its relative B -derivative proves commutation at the right.

Consequently the source's two-term-complex comparison and six-term sequence have exactly the required conormal row (C4). More explicitly, the connecting map takes a class in the kernel of the conormal differential for J , lifts its equation to K , differentiates in T/A , and retains the dx_s component. Its dy_t component is zero by the cycle condition. Its class in $\Omega_{B/A} \otimes_B C$ is unchanged by changing the lift: the difference lies in IT , and its derivative is a boundary from the left member of (C4). A change by an element of J^2 can be lifted to K^2 , whose derivative vanishes after tensoring with C . The resulting kernel/image equalities follow directly: a middle cycle mapping to zero in the right kernel is a left cycle by (C4) and (C5); a right cycle whose connecting class is zero can have its lift corrected by a left conormal element to become a middle cycle; and a left differential class dying in the middle cokernel is the connecting image of the right image of an element whose middle boundary represents it. Exactness at the other cokernel positions is right exactness of (C4) and (C5). The initial kernel map is injective because the initial conormal map is injective. These are the original source maps, with the tensor over B .

The wrong subscript has an explicit counterexample satisfying the source hypothesis. Let

$$A = k, \quad P = k[x], \quad I = K = (x^2), \quad B = C = k[x]/(x^2),$$

and take the identity $B \rightarrow C$, presented with no y -variables. Its kernel $J = 0$ is generated by the empty Koszul-regular sequence, so this map is a local complete intersection. The source quotient is

$$IT/IK = (x^2)/(x^4) = I/I^2,$$

with k -basis $u = x^2 \bmod x^4$, $v = x^3 \bmod x^4$. In contrast, $(I/I^2) \otimes_k C$ has the four-element basis

$$u \otimes 1, \quad u \otimes \bar{x}, \quad v \otimes 1, \quad v \otimes \bar{x}.$$

Its multiplication map to I/I^2 sends these four basis vectors to $u, v, v, 0$, respectively. Its kernel has the independent basis

$$u \otimes \bar{x} - v \otimes 1, \quad v \otimes \bar{x}.$$

Indeed a vector with coefficients a, b, c, d maps to $au + (b+c)v$, so it is in the kernel exactly when $a = 0, c = -b$. This proves the erroneous tensor cannot be identified with the source quotient and cannot supply the asserted injection. The B -balanced tensor identifies the middle two basis vectors and makes the last zero, exactly as (C1) requires.

Propagation: replace the two printed A subscripts by B . The diagram, the six-term statement, and the existing Algebra cotangent calculations already use the correct B -action. The exact identification (C1) needs no local-complete-intersection assumption; the injection into K/K^2 uses the explicitly proved intersection equality. This distinction is a proved dependency fact, not a proposed new theorem attributed to the source authors.

CC-SERIES-FACTOR: retain the coefficient's factor of x^{-1}

Source: Examples, the original series and the localized comparison, lines 1428–1483. The faulty recurrence is at line 1481.

Keep the source field k , the original indeterminate x , and all coefficients of its stipulated transcendental series

$$z = \sum_{i=1}^{\infty} a_i x^i \in k[[x]], \quad z_j = x^{-j} \left(z - \sum_{i=1}^{j-1} a_i x^i \right) = \sum_{i=j}^{\infty} a_i x^{i-j} \quad (j \geq 1).$$

The coefficient calculation in the original Laurent series field gives

$$xz_{j+1} = \sum_{i=j+1}^{\infty} a_i x^{i-j} = z_j - a_j, \quad z_{j+1} = x^{-1}(z_j - a_j). \quad (S1)$$

The printed expression $x^{-1}z_j - a_j$ minus the right side of (S1) is $a_j(x^{-1} - 1)$. The series is transcendental over $k(x)$, so at least one of its coefficients is nonzero. At any such index this difference is nonzero in $k((x))$. Neither the coefficient nor the power of x can be discarded.

Let $R = k[x, z_1, z_2, \dots] \subset k[[x]]$, as in the source. Its Laurent localization satisfies the exact equality of subrings of $k((x))$

$$R[1/x] = k[x, x^{-1}, z]. \quad (S2)$$

The inclusion from right to left uses $z = xz_1 \in R$ and the adjoined inverse of x . For the other inclusion, the full formula for every z_j displays it in the right-hand ring. Thus both maps are induced by the same original elements in $k((x))$.

We also verify the residue ideal used by the localized comparison. Each z_j is transcendental over $k(x)$: an algebraic relation for z_j , combined with $z = x^j z_j + \sum_{i < j} a_i x^i$, would make z algebraic. Hence $R_j = k[x, z_j]$ is the polynomial ring on the two displayed elements. The exact transition maps

$$R_j \rightarrow R_{j+1}, \quad x \mapsto x, \quad z_j \mapsto xz_{j+1} + a_j$$

are inclusions in $k[[x]]$, and their union is R . Modulo $x - 1$ they become the polynomial-ring isomorphisms

$$k[z_j] \rightarrow k[z_{j+1}], \quad z_j \mapsto z_{j+1} + a_j.$$

Consequently

$$R/(x - 1) \xrightarrow{\sim} k[z], \quad x \mapsto 1, \quad z_j \mapsto z - \sum_{i=1}^{j-1} a_i. \quad (S3)$$

This can be checked without assuming quotient-commutation: define the compatible maps on each R_j by these formulas; they induce the map on the union R . Its inverse sends the polynomial indeterminate z to the class of the original xz_1 . The transition relations verify both composites on every generator. Thus its kernel is exactly $(x - 1)$.

The source ideal

$$\mathfrak{n} = (x - 1, z_1, z_2 + a_1, z_3 + a_1 + a_2, \dots)$$

is therefore $(x - 1, z)$, with residue field k . In particular $x \notin \mathfrak{n}$. Equality (S2) gives the actual comparison

$$k[x, x^{-1}, z]_{(x-1, z)} \xrightarrow{\sim} R_{\mathfrak{n}}. \quad (S4)$$

The forward map sends the original x, x^{-1}, z to those same elements. For its inverse, send each z_j to

$$x^{-j} \left(z - \sum_{i=1}^{j-1} a_i x^i \right).$$

These formulas respect every ring relation because (S2) identifies the two rings inside $k((x))$. An element of $R \setminus \mathfrak{n}$ maps to an element whose value at $x = 1, z = 0$ is nonzero by (S3), and hence

is invertible in the left side of (S4). Conversely a denominator outside $(x - 1, z)$ maps outside the maximal ideal on the right. Both maps therefore extend to the indicated localizations, and their composites are the identity on all numerators and denominators. This proves the source's localized comparison with every original coefficient retained.

Propagation: the correction changes the recurrence used to display the inverse in (S4), while the original ring, prime ideal and localization map remain the same. The receiving Algebra descent note already uses (S1); its finite étale construction does not require a new source object. Its complete argument remains a separate editorial note. This check does not claim to re-review every assertion of the Examples chapter.

Integration scope

The preparation records must retain each original statement, the four exact source operations, both original chapter files, the prior complete cumulative chapter files, source/proof locators, and the receiving Algebra group IDs. Removing these four edits must recover the complete prior chapters, including their existing additions. The changes must also produce standalone fixes-only patches against the original authority, with no unrelated theorem additions.

This source-ordered propagation does not make the Algebra R58 candidate or either cross-chapter correction public or admitted. Those transitions require their own source replay, build evidence, rendered checks, and publication receipts.

2 Brauer Groups: source review and propagation of the received corrections

Source: The Stacks Project authors, brauertex, authority commit a04446e57ec1fbc252a871afcec7752fb2807b14, SHA-256 B2504820D769EBE4E9E33B8ADD78753FB30ACA6E8A7F75C8D54DDA885EDCD682. The retained original TeX has 817 lines; all were read for this review. The twelve received reports and ten existing source operations are retained separately. This note supplies editorial arguments and propagation findings. It does not silently replace the source or claim a new research theorem.

1. The module and multiplication conventions

The source defines modules to be unital **right** modules at lines 33–37. In its Rieffel proof, lines 100–106, the endomorphisms of a right module act on the left, whereas endomorphisms of the resulting left module are made to act on the right. Its footnote explicitly reverses multiplication in the latter ring. We retain both conventions and distinguish their maps.

Write End^{comp} for endomorphisms multiplied by composition, $fg = f \circ g$. Write $\text{End}^{\text{right}}$ for the same functions with multiplication $f \star g = g \circ f$. The identity on functions is an isomorphism $\text{End}^{\text{right}} \cong (\text{End}^{\text{comp}})^{\text{op}}$, not an isomorphism to End^{comp} in general. Thus the source's first commutant is $\text{End}_A^{\text{comp}}(M)$, acting on the left, and its bicommutant is $\text{End}_L^{\text{right}}(M)$, acting on the right.

For the right action of any ring A , define $R_a(m) = ma$. Then

$$\begin{aligned}(R_a \circ R_b)(m) &= (mb)a = m(ba) = R_{ba}(m), \\ (R_a \star R_b)(m) &= (ma)b = m(ab) = R_{ab}(m).\end{aligned}$$

Consequently the original action is a homomorphism into $\text{End}^{\text{right}}$ and an antihomomorphism into End^{comp} . No opposite algebra can be removed or introduced without checking this multiplication.

2. Matrix corners and the two inverse module maps

This checks MC-STK-ERR-0745 against source lines 259–278. Let R be the original possibly noncommutative ring, let $n \geq 1$, and put $R_n = \text{Mat}(n \times n, R)$. Denote the matrix units by e_{ij} ; for $r \in R$, re_{ij} denotes the matrix with r in position (i, j) and zero elsewhere. The multiplication formula, with the factors in their original order, is

$$(re_{ij})(se_{uv}) = \delta_{ju}(rs)e_{iv}.$$

Let J be a two-sided ideal of R_n . The additive sum $J = \bigoplus_{1 \leq i, j \leq n} e_{ii} J e_{jj}$ is direct because its summands have disjoint matrix positions. Put $I = \{r \in R : re_{11} \in J\}$. Addition, additive inverses, and multiplication on either side by se_{11} prove that I is a two-sided ideal of R . For every pair i, j , the two identities

$$\begin{aligned}e_{i1}(re_{11})e_{1j} &= re_{ij}, \\ e_{1i}(re_{ij})e_{j1} &= re_{11}\end{aligned}$$

prove $e_{ii} J e_{jj} = \{re_{ij} : r \in I\}$. The inverse maps between this corner and I take r to re_{ij} and extract its (i, j) coefficient. They do not identify different corners as subsets of the matrix ring. Summing the corners proves $J = IR_n$. The retained source correction expresses precisely this common coefficient ideal; its unexpanded direct-sum notation has the unique available pair of indices and is not another defect.

For completeness, the source's equivalence of right-module categories is given by actual inverse maps. Regard $X^{\oplus n}$ as rows over a right R -module X , with $(xU)_j = \sum_i x_i u_{ij}$. Conversely take Ye_{11} for a right R_n -module Y , with its R -action through $r \mapsto re_{11}$. Define

$$\begin{aligned}\alpha_Y : (Ye_{11})^{\oplus n} &\rightarrow Y, & (y_i)_i &\mapsto \sum_i y_i e_{1i}, \\ \beta_Y : Y &\rightarrow (Ye_{11})^{\oplus n}, & y &\mapsto (ye_{i1})_i.\end{aligned}$$

The entries of β lie in Ye_{11} since $e_{i1}e_{11} = e_{i1}$. The relations $e_{1j}e_{i1} = \delta_{ji}e_{11}$ and $\sum_i e_{i1}e_{1i} = \sum_i e_{ii} = 1$ give $\beta_Y\alpha_Y = \text{id}$ and $\alpha_Y\beta_Y = \text{id}$. Matrix multiplication verifies that these are right R_n -module maps. The analogous identification $(X^{\oplus n})_{e_{11}} \rightarrow X$ extracts the first entry and its inverse inserts zeros. All these maps commute with module homomorphisms, so they give the stated equivalence, including for modules which are not finitely generated.

Commutation with every e_{ii} forces a central matrix to be diagonal; commutation with every e_{ij} makes its diagonal entries equal. Commutation with re_{11} for every r then forces that common entry into $Z(R)$. Conversely each such scalar matrix is central. This proves the exact center map $Z(R) \rightarrow Z(R_n), r \mapsto r \cdot 1_n$, with inverse taking entry $(1, 1)$.

3. The zero-algebra correction does not yet reach every dependent statement

The ten existing corrections include MC-STK-ERR-0746, requiring a simple algebra to be nonzero at source lines 63–64. This is necessary: the zero algebra satisfies the printed two-sided-ideal condition, but cannot be a positive-size matrix algebra over a skew field, whose identity is nonzero.

However, the separate lemma at lines 119–138 assumes only that A is finite, not that A is simple. Its assertion (1), that A has a simple module, is still false for $A=0$ after the definition of “simple algebra” is repaired. For a unital module over the zero algebra, every m satisfies $m = m \cdot 1 = m \cdot 0 = 0$. Therefore all such modules are zero and none is simple.

The precise correction restricts **assertion (1)** to $A \neq 0$; assertions (2)–(4) remain valid for every finite algebra, including zero. Their full argument is as follows. In any nonzero right A -module N , choose $m \neq 0$. Then $m = m1$ belongs to the cyclic submodule mA , which is a nonzero quotient of the finite-dimensional k -vector space A . Among nonzero submodules of mA choose one of least positive k -dimension, called M . Any nonzero proper submodule of M would have smaller positive dimension, so M is simple. This proves (2). Applying (2) to the module A proves (1) when A is nonzero. Conversely existence of a simple unital A -module implies A is nonzero by the preceding zero-algebra calculation.

If M is simple, any $m \neq 0$ generates it, since $0 \neq mA \subset M$. Hence M is a quotient of A and has finite k -dimension, proving (3). Finally a nonzero A -endomorphism f of M has kernel zero and image M , since its kernel and image are submodules. It is bijective, and its inverse is A -linear: if $f(x) = m$, then $f(xa) = ma$, so $f^{-1}(ma) = f^{-1}(m)a$. The identity of M is nonzero. Thus $\text{End}_A^{\text{comp}}(M)$ is a skew field, proving (4). This last argument requires no finiteness assumption at all; the general Schur assertion is a proved underclaim recorded editorially.

The word “nonzero” is also needed before “submodule of minimal (finite) dimension” in the proof at line 133: among all submodules, zero has the smallest dimension. This is a bounded repair of the stated minimization.

Downstream: the construction of a simple submodule of a finite simple A in Wedderburn’s proof is now valid because the corrected definition already gives $A \neq 0$. No new hypothesis is needed in that theorem or in its later uses. The unrelated zero-module exception in assertion (6) at lines 295–296 is correctly excluded by existing MC-STK-ERR-0747.

4. The exact endomorphism rings of the original right matrix module

Retain the source’s $A = \text{Mat}(n \times n, K)$ and its right module $M = K^{\oplus n}$, with row action. For each $a \in K$ let $L_a(v_1, \dots, v_n) = (av_1, \dots, av_n)$. Associativity shows $L_a(vU) = L_a(v)U$, so L_a is A -linear. Moreover

$$L_a \circ L_b = L_{ab}.$$

Every A-linear f is of this form. If ε_i is the i -th unit row, then $\varepsilon_i e_{ii} = \varepsilon_i$ implies $f(\varepsilon_i)$ has only its i -th entry, say a_i . The equation $\varepsilon_i e_{ij} = \varepsilon_j$ gives $a_i = a_j$ for every pair. Write their common value a . Using the matrix with r in diagonal position (i, i) gives $f(r\varepsilon_i) = ar\varepsilon_i$. Additivity now gives $f = L_a$ on every row. Evaluation on ε_1 supplies the inverse to $a \mapsto L_a$. Thus the exact ring isomorphism is

$$K \rightarrow \text{End}_A^{\text{comp}}(M), \quad a \mapsto L_a,$$

and its target is K , not K^{op} , under the source's left action convention. The formula is compatible with the central k -actions on both sides.

Now let M be a left K -vector space with ordered basis $\varepsilon_1, \dots, \varepsilon_n$, retaining coefficients on the left. A left K -linear map is determined by $t(\varepsilon_i) = \sum_j u_{ij}\varepsilon_j$ and sends the coefficient row v to vU . Let t_U denote this map. Direct multiplication gives

$$t_U \circ t_V = t_{VU}, \quad t_U \star t_V = t_{UV}.$$

Consequently $\text{End}_K^{\text{right}}(M) \cong \text{Mat}(n \times n, K)$ by $t_U \mapsto U$, whereas $\text{End}_K^{\text{comp}}(M) \cong \text{Mat}(n \times n, K^{\text{op}})$ by $t_U \mapsto (u_{ji}^{\text{op}})_{i,j}$. The latter transpose retains every entry and reverses multiplication inside K ; it is not a change in the original action.

This proves that the final line 158 of the source's Wedderburn proof, which uses its explicitly defined right-acting bicommutant, should have $\text{Mat}(n \times n, K)$, with the original $K = \text{End}_A^{\text{comp}}(M)$. Likewise the expressions $\text{End}_A(M) = K^{\text{op}}$ at lines 290 and 411 and $L = K^{\text{op}}$ at line 311 are inconsistent with that same convention. The correct identifications there use K . The uniqueness proof at line 413 then directly gives $K \cong K'$. The theorem that a finite simple algebra is some matrix algebra over a skew field remains true; the error is in the explicitly named division algebra and its action, not in the existential classification.

The order discrepancy is substantive even without finding a division algebra abstractly nonisomorphic to its opposite. As an explicit witness, inside $\text{Mat}(2 \times 2, C)$ put

$$I = \text{diag}(i, -i), \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Q = IJ.$$

Their real span with the identity is an associative real algebra because it is closed under multiplication, with $I^2 = J^2 = -1$ and $JI = -IJ$. The four matrices are linearly independent over $\mathbb{R}$. For $h = a + bI + cJ + dQ$, multiplication by $\bar{h} = a - bI - cJ - dQ$ in either order gives $(a^2 + b^2 + c^2 + d^2) \cdot 1$. Every nonzero h is therefore invertible. In this skew field, $L_I \circ L_J = L_Q$, while multiplication of I and J in the opposite ring gives $JI = -Q$, which would map to $-L_Q$. These are unequal because $2Q \neq 0$. Thus the source's unmodified left action cannot identify its composition ring with K^{op} by the claimed coefficient assignment.

5. Bicommutants, including the missed generality for the module

For the finite case in source lines 280–317 write $A = \text{Mat}(n \times n, K)$, $M = K^{\oplus n}$, and $N = M^{\oplus r}$, where r is the actual number of simple summands. For $r \geq 1$, an A-linear endomorphism of N is a matrix $(b_{ij}) \in \text{Mat}(r \times r, K)$ acting by $(v_j)_j \mapsto (\sum_j b_{ij}v_j)_i$. The calculation in Section 4 proves both that every endomorphism has these entries and that composition is ordinary matrix multiplication. Hence

$$B = \text{End}_A^{\text{comp}}(N) \cong \text{Mat}(r \times r, K) = \text{Mat}(r \times r, L), \\ L = \text{End}_A^{\text{comp}}(M) \cong K.$$

The right action of A identifies A with $\text{End}_B^{\text{right}}(N)$. Here is a proof which also establishes the exact stronger statement for **every nonzero right A-module N** , without requiring N finite.

The inverse equivalences in Section 2, with $R=K$, identify N with $(Ne_{11})^{\oplus n}$. Choose a basis of the right K -vector space Ne_{11} . Writing its index set as T gives an A-module isomorphism $N \cong \bigoplus_{t \in T} M_t$,

where each $M_t = M$, and every element has finite support. The set T is nonempty because N is nonzero. This is an explicit direct sum of the original simple right matrix modules, with no finiteness restriction on T .

Put $B = \text{End}_A^{\text{comp}}(N)$. Suppose F is an additive map commuting with all of B . For each t , the projection onto M_t belongs to B , so F preserves each M_t . The A -linear map which transfers the row in M_t to M_u and annihilates the other summands also belongs to B . Commutation with these maps shows that the restrictions of F are one and the same additive map $f : M \rightarrow M$, under the displayed identifications. For every $a \in K$, the map which applies L_a to a single summand and is zero on all others belongs to B . Therefore $f(av) = af(v)$: f is left K -linear. Section 4 now gives a unique matrix $U \in \text{Mat}(n \times n, K)$ with $f(v) = vU$. On an arbitrary finite-support element (v_t) , additivity and the projections give $F((v_t)) = (v_t U)$. Thus F is exactly the original right action R_U on N . Conversely every R_U commutes with every A -linear map by the definition of A -linearity. The action is faithful: if R_U is zero, apply it to each unit row in any one nonempty summand to read off all entries of U . The multiplication calculation in Section 1 proves

$$A \cong \text{End}_B^{\text{right}}(N), \quad A^{\text{op}} \cong \text{End}_B^{\text{comp}}(N).$$

For $N=0$ the target is the zero ring, so neither can recover nonzero A . This proves both the necessity of the existing nonzero correction and the editorial underclaim: finiteness of N is unnecessary for the bicommutant conclusion. It is still needed for the accompanying description of B as a finite-size matrix algebra. No change to that finite-matrix assertion is inferred from the stronger bicommutant conclusion.

The dimensions in assertion (5) remain exactly $[A : k] = n^2[K : k]$, $[L : k] = [K : k]$, $\dim_k(M) = n[K : k]$, and hence $[A : k][L : k] = \dim_k(M)^2$. The centers of A and L identify with $Z(K)$ by the explicit scalar-matrix map of Section 2.

6. Propagation through Skolem–Noether and the centralizer argument

This checks the mathematical qualification in MC-STK-ERR-0748 and the explicit multiplication map in MC-STK-ERR-0749.

In the source Skolem–Noether proof retain the right A -module M and $L = \text{End}_A^{\text{comp}}(M)$, acting on its left. For embeddings $f, g : B \rightarrow A$, the two right actions of $H = B \otimes_k L^{\text{op}}$ are

$$m \cdot_1 (b \otimes l^{\text{op}}) = lmf(b), \quad m \cdot_2 (b \otimes l^{\text{op}}) = lmg(b).$$

For example, applying $(b_1 \otimes l_1^{\text{op}})$ and then $(b_2 \otimes l_2^{\text{op}})$ gives $l_2 l_1 m f(b_1) f(b_2)$, exactly the action of their product in H . Their k -dimensions are equal. The source's finite simple classification therefore supplies an H -module isomorphism φ from the first action to the second. Its L -linearity identifies it with R_x for $x \in A$ by the bicommutant result. Its inverse is R_y for $y \in A$, so faithfulness gives $xy = yx = 1$. Intertwining the B -actions gives $f(b)x = xg(b)$, hence $f(b) = xg(b)x^{-1}$. The opposite on L is essential and stays in this proof.

Taking $B=A$, f a k -algebra automorphism, and g the identity proves the corrected inner-automorphism assertion. Without “ k -algebra”, take $k=\mathbb{C}$, $A=\mathbb{C}$, and complex conjugation. It is a unital ring automorphism which is not the identity. Every inner automorphism of the commutative field $\mathbb{C}$ is the identity, so the unqualified ring assertion would be false.

For the centralizer theorem retain the source's $H = B \otimes_k L^{\text{op}}$ acting on the right of M and $C = \text{Cent}_A(B)$. The assignment $c \mapsto R_c$ takes C into $\text{End}_H^{\text{comp}}(M)$. It is bijective: an H -linear map commutes with L , so the A -bicommutant identifies it with some R_a ; commutation with the B -action says $ba = ab$ for every b , by faithfulness of M . Thus $a \in C$. But composition is reversed: $R_c \circ R_d = R_{dc}$. The exact identification in source line 546 is therefore

$$C^{\text{op}} \cong \text{End}_H^{\text{comp}}(M).$$

If $H = \text{Mat}(m \times m, K)$ and M is a direct sum of n copies of its right simple module $K^{\oplus m}$, Section 5 gives $\text{End}_H^{\text{comp}}(M) \cong \text{Mat}(n \times n, K)$. Taking opposites and the entrywise transpose gives $C \cong \text{Mat}(n \times n, K^{\text{op}})$. **The opposite in source line 553 is correct and must be retained.** It differs from the incorrect occurrences in the right simple module's left-acting endomorphism ring. All original dimension formulas at lines 556–558 remain unchanged:

$$\begin{aligned} \dim_k(M) &= nm[K : k], & [B : k][L : k] &= m^2[K : k], \\ [C : k] &= n^2[K : k], & [A : k][L : k] &= \dim_k(M)^2. \end{aligned}$$

Since $[L : k] > 0$, substitution and cancellation of that nonzero integer give $[A : k] = [B : k][C : k]$. The source double-centralizer conclusion follows from $B \subset \text{Cent}_A(C)$ and applying this formula to C ; equality of the finite k -dimensions makes the inclusion surjective.

When B is central, $Z(C) = C \cap \text{Cent}_A(C) = C \cap B = Z(B) = k$. The map in the existing corrected tensor statement is

$$\mu : B \otimes_k C \rightarrow A, \quad \mu(b \otimes c) = bc.$$

It is well defined by k -balance. It is multiplicative since C commutes with B : $(bc)(b'c') = bb'cc'$. It takes the identity to the nonzero identity of A . Its domain is simple by the source tensor-simplicity lemma, so its kernel is zero. Its domain and codomain have the same finite dimension by the formula just proved, so μ is an isomorphism. This supplies the exact map rather than treating distinct presentations as unrelated. The source's equality was usable as a customary canonical identification; the existing edit is recorded as a clarification making that map explicit, not as a new counterexample to the theorem.

7. The corrected separability negation and its receiving argument

Both producer reports at source line 733 refer to existing MC-STK-ERR-0017. The sought element at lines 724–726 belongs to $K \setminus k$ and is separable over k . Its negation is that no element of $K \setminus k$ is separable, not that K has no separable element at all. Every $a \in k$ has minimal polynomial $T - a$, whose derivative is 1, so the latter statement is always false. Existing MC-STK-ERR-0016 independently repairs the spelling of “splitting field” at line 711. These are two shared mathematical loci, each reported twice; they are not four distinct corrections.

The corrected negation really does give the next assertion in the proof. In characteristic $p > 0$, let f be the monic irreducible polynomial of $x \in K$. Repeatedly remove powers of p from every exponent until $f(T) = g(T^{p^r})$ with $g' \neq 0$. The polynomial g is irreducible, since a factorization of g would give one of f . Therefore it is separable, and $y = x^{p^r}$ is separable over k . The corrected assumption forces $y \in k$. Since g is the monic irreducible polynomial of y , $g(U) = U - y$; thus $f(T) = T^{p^r} - y$. Its degree p^r is at most $[K : k]$. Choose a p -power q at least $[K : k]$. Each such p^r divides q , hence $x^q \in k$ for every x .

With the original basis $a_1 = 1, a_2, \dots, a_n$, write every ordered product $a_{i_1} \dots a_{i_q} = \sum_j c_{i_1, \dots, i_q}^j a_j$, keeping all q factors. Then the coordinate polynomials in the source are exactly

$$f_j(X_1, \dots, X_n) = \sum_{1 \leq i_1, \dots, i_q \leq n} c_{i_1, \dots, i_q}^j X_{i_1} \dots X_{i_q}.$$

For $j \geq 2$ they vanish on all k^n . Over the infinite field k this implies they are the zero polynomials: induct on the number of variables, using that a nonzero one-variable polynomial has at most its degree many roots and applying that assertion to each coefficient. Their vanishing therefore persists over the algebraic closure. The resulting matrix algebra has size at least 2, since $K \neq k$ and $[K : k] = d^2$. Its element e_{11} has $e_{11}^q = e_{11}$, and is not central because $e_{11}e_{12} = e_{12}$ whereas $e_{12}e_{11} = 0$. This is the required contradiction.

The existing punctuation edits at lines 561 and 728 and article insertion at line 771 preserve all mathematical assertions. No change to those already-admitted operations is required by this review.

8. Scope and remaining integration

The ten existing corrections replay exactly to the current complete brauer.tex; all twelve received reports have been compared with their actual source operations. The separate zero-algebra lemma and the endomorphism-order propagation above are new findings from this review. Their bounded edits and this complete editorial argument require their own prepared source record and later admission. Neither reading the chapter nor proving these comparisons is a publication receipt.

The stronger Schur and nonzero-module bicommutant conclusions are proved editorial under-claims. They do not authorize replacing the source's finite-module exposition. The later joint synthesis and novelty review remain after the core correction work. No claim about the separately received open Brauer problem or its proposed solution is made here.